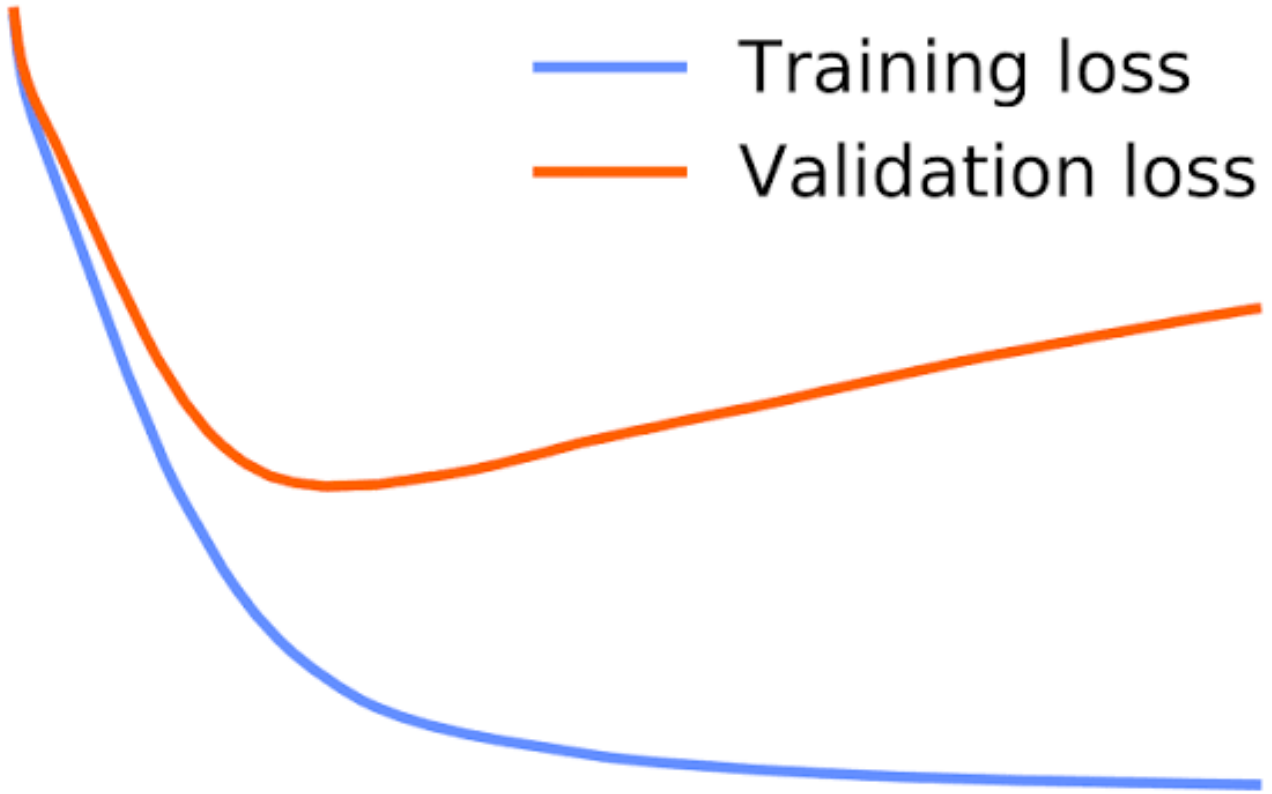
Machine Learning

Week 1: Introduction & regression



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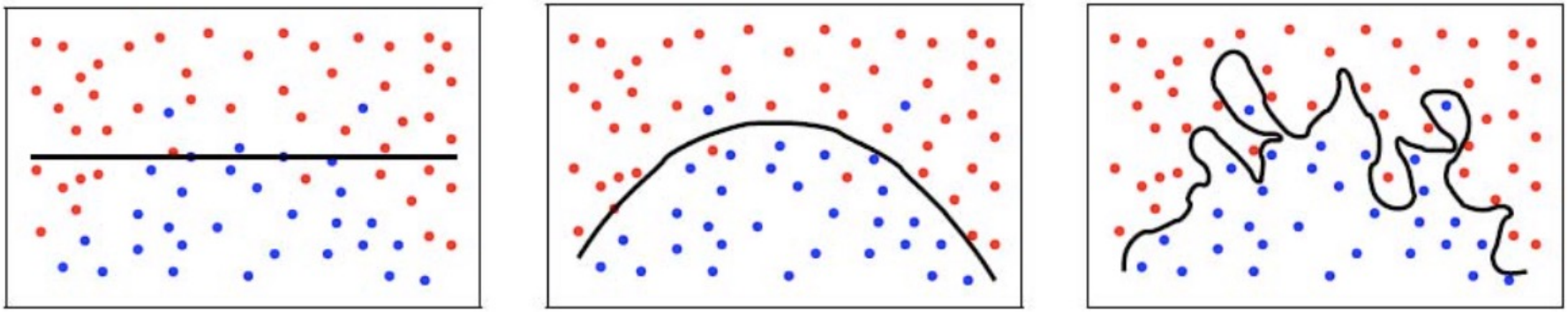
Overfitting, in images

Classification

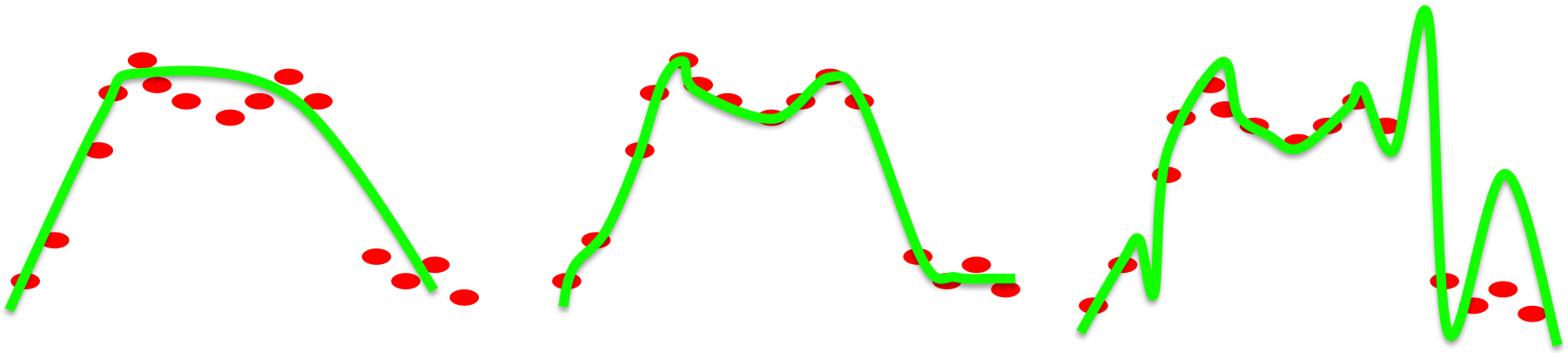
Underfitting

just right ⟷

Overfitting



Regression



Ridge regression

$$L_{\text{ridge}}(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_2^2$$

Take the gradient with respect to w

$$\nabla_{\mathbf{w}} L_{\text{ridge}}(\mathbf{w}) = -2X^T(\mathbf{y} - X\mathbf{w}) + 2\lambda\mathbf{w}$$

Set the gradient to zero and rearrange

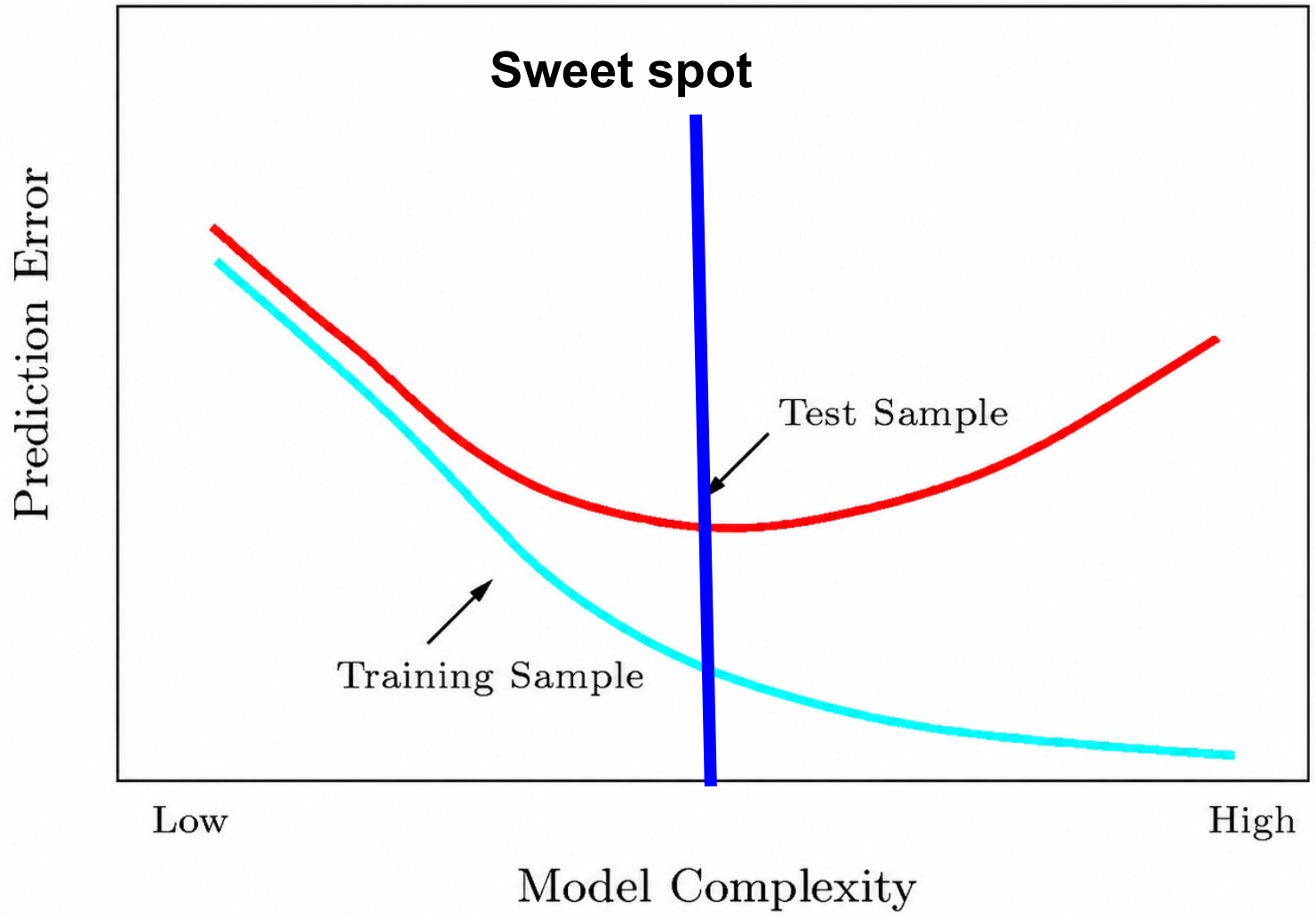
$$(X^T X + \lambda I)\hat{\mathbf{w}} = X^T\mathbf{y}$$

I is the identity matrix. For λ > 0, the solution is unique.

$$\hat{\mathbf{w}} = (X^T X + \lambda I)^{-1} X^T\mathbf{y}$$

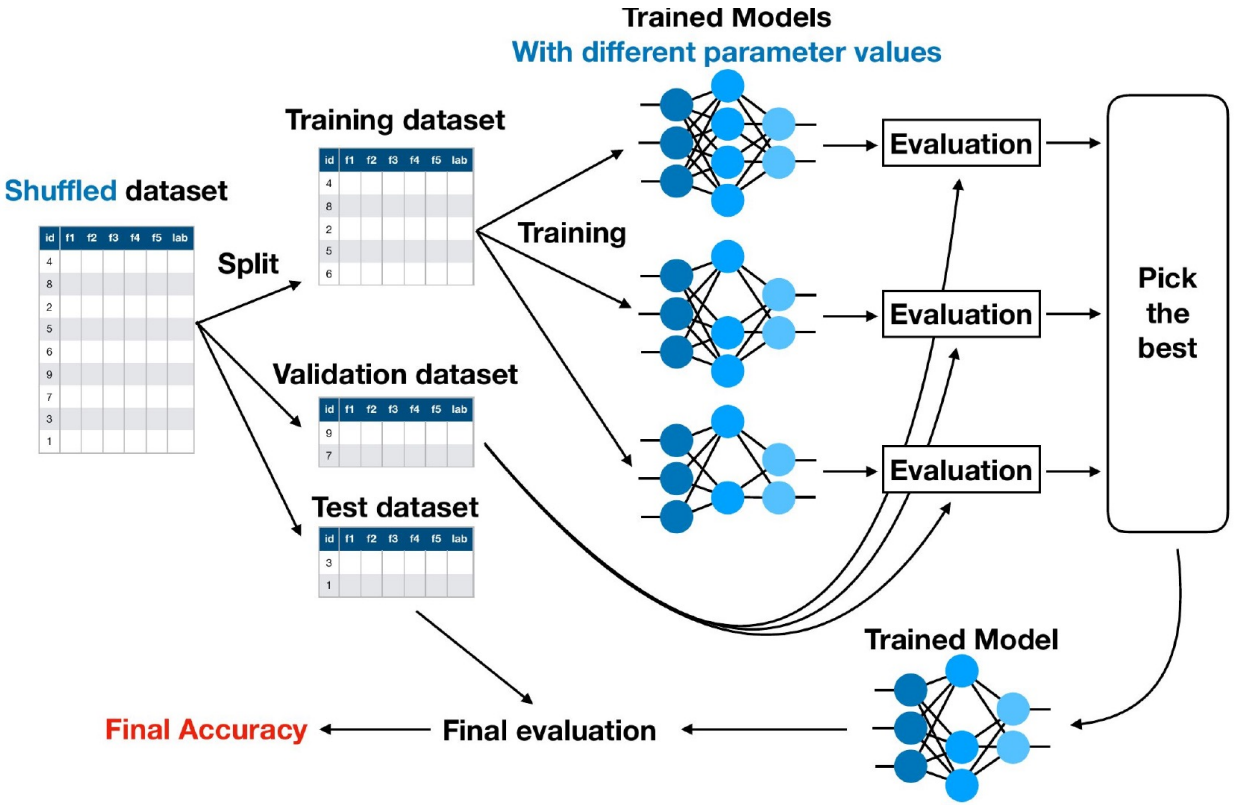
In code, solve the linear system.

Model complexity and held-out error



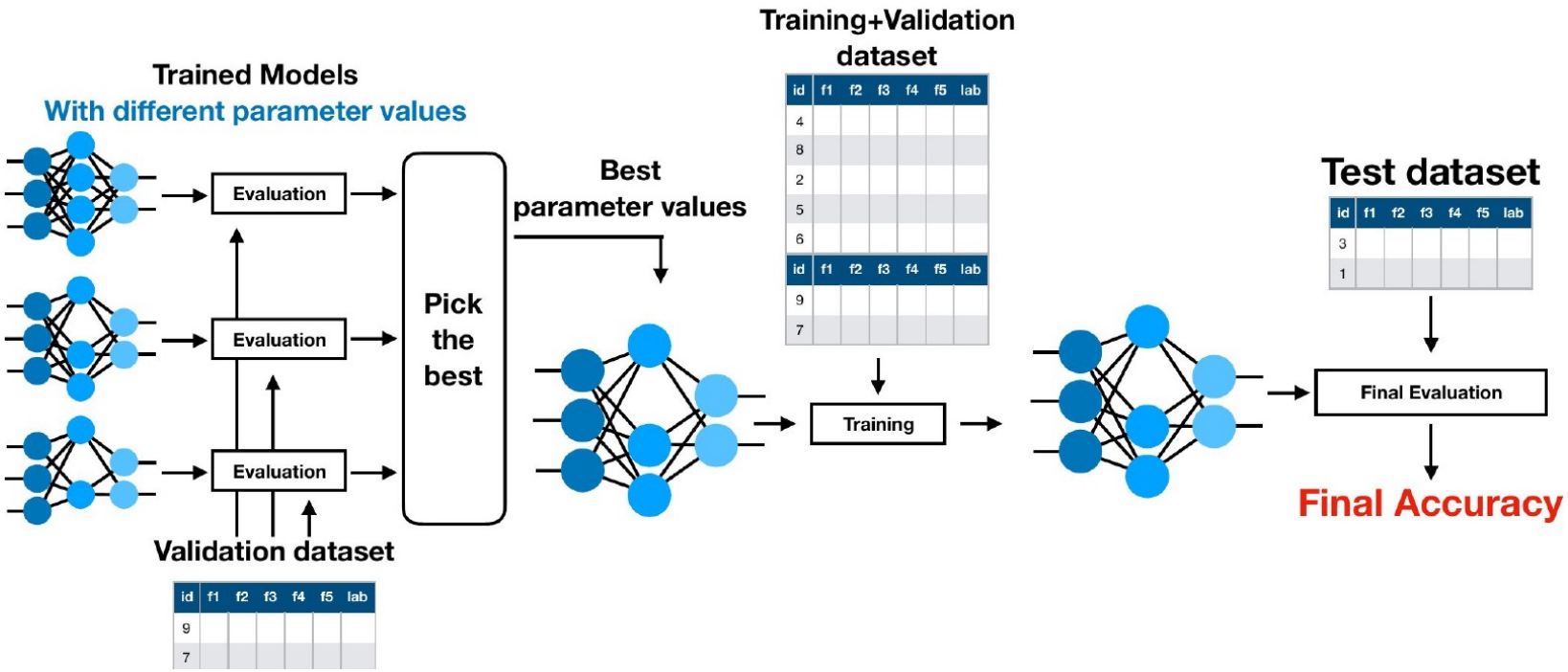
Larger ridge λ means lower complexity.

Training, validation and test data



- Fit model parameters on training data and select hyperparameters using validation data.
- Reserve the test set for evaluating the final modelling procedure.

Final evaluation



- After choosing hyperparameters, optionally refit on training and validation data together.
- Evaluate once on the untouched test set. Further choices need fresh validation.

Standardise features before regularising

Feature j: training mean μ_j and standard deviation $s_j > 0$.

$$\widetilde{x}_j = \frac{x_j - \mu_j}{s_j}$$

Unit variance gives each feature a comparable scale.

Same predictions, different ridge penalty

Scale feature x_j by a nonzero factor c and its coefficient w_j by 1/c.

$$x_j \mapsto c x_j, \qquad w_j \mapsto w_j / c, \qquad \lambda w_j^2 \mapsto \lambda w_j^2 / c^2$$

100× larger feature: penalty ÷ 10,000
0.1× feature: penalty × 100

The intercept has regulariser weight zero

$$\text{Penalty} = \lambda \sum_{j=1}^{D} w_j^2 \qquad (\text{no } w_0^2 \text{ term})$$

w_0: intercept
D: number of features

Choose a split that matches deployment

Independent examples → rotate the held-out fold

K-fold cross-validation divides the data into K folds. Train on K − 1 folds, validate on the remaining fold, and rotate.

New people, devices or sites → hold out whole groups

Keep related observations in the same fold. A row split can test on a device that the model has already seen.

Future observations → hold out later times

Train on earlier data and validate on later data. Choose the split to match the prediction you will make in the real world.

Keep validation information out of training

Fit preprocessing inside each training fold

Estimate means, standard deviations and any feature selection using that fold's training examples only. Apply the fitted transformations unchanged to its validation examples.

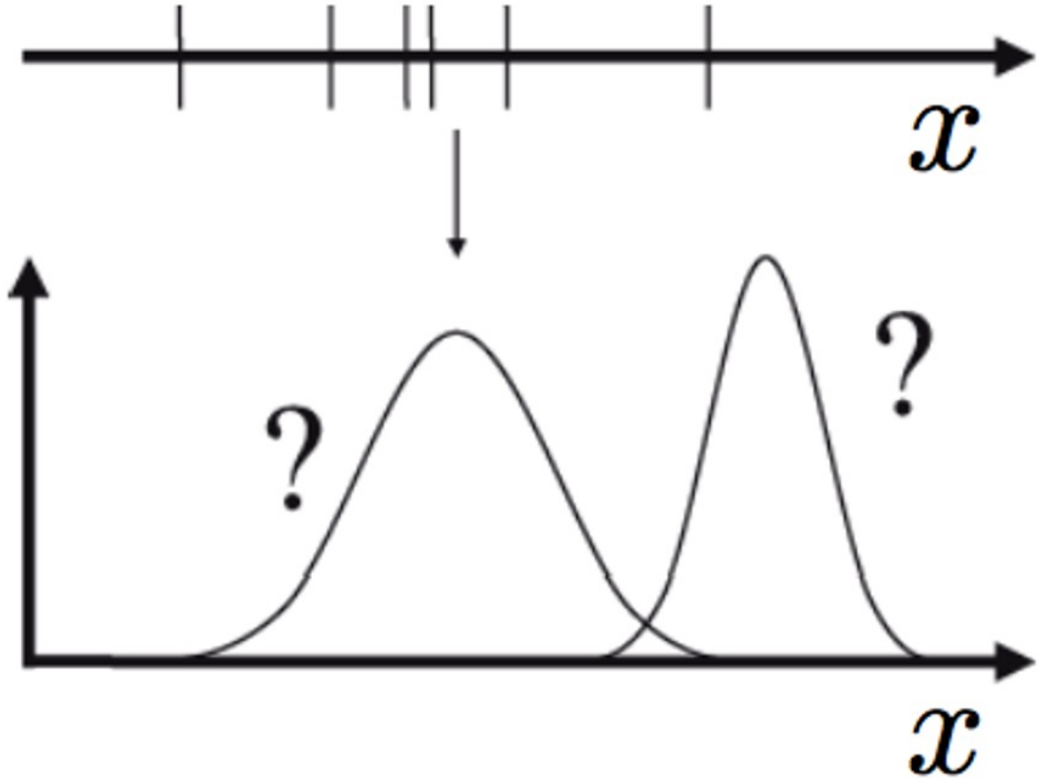
Example: many sensor readings from each device

Will deployment involve known devices or new devices?

For new devices, reserve entire devices for validation. Randomly splitting their readings leaks device-specific information across the split.

The validation score should reflect what the model can do with genuinely unseen information.

Probabilistic modeling



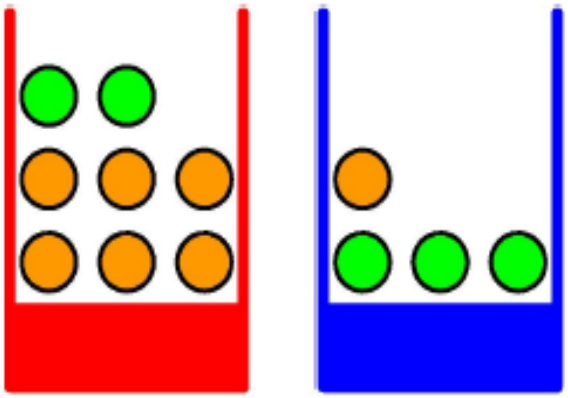
Murphy (2022), Foundations

Random variables and observed values

Y is a random variable; y is one observed value.

Example: choose a box, then a fruit

Y records the fruit: apple (a) or orange (o).
B records the box: red (r) or blue (b).



$$Y \in \{a, o\}, \qquad y = a \text{ or } y = o$$

$$p(y) = p(Y = y) = \Pr(Y = y)$$

$$p(y) \geq 0, \qquad \sum_y p(y) = 1$$

p(y) and Pr(Y = y) denote probability. PMF: probability mass function.

Joint, marginal and conditional probability

Choose red with probability 0.4, blue with probability 0.6.

Joint probability	Red box	Blue box	Fruit total
Apple	0.10	0.45	0.55
Orange	0.30	0.15	0.45
Box total	0.40	0.60	1.00

$$P(Y = a, B = r) = 0.10 \qquad P(B = r) = 0.10 + 0.30 = 0.40$$

Conditional probability

$$P(Y = y, B = b) = P(Y = y \mid B = b)\, P(B = b)$$

The bar | means “given”. Assume P(B = b) > 0.

$$P(Y = a \mid B = r) = \frac{0.10}{0.40} = 0.25$$

Normalise the red column by its total.

Bayes' rule: reverse the conditioning

Observed: an apple. Which box did it come from?
B: box colour; r: red. Y: fruit; a: apple.

$$p(B = b \mid Y = a) = \frac{p(Y = a \mid B = b)\, p(B = b)}{p(Y = a)}$$

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

Prior: before observing fruit. Posterior: after observing it.
Likelihood: apple chance in a chosen box. Evidence: total apple chance.

$$p(B = r \mid Y = a) = \frac{0.25 \times 0.40}{0.55} = \frac{2}{11} \simeq 0.182$$

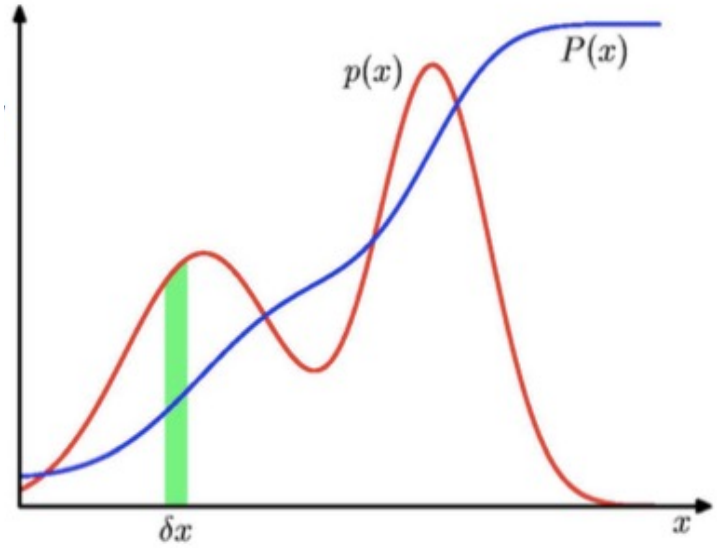
An apple makes the blue box more likely.

Continuous variables: density and probability

X is a continuous random variable. Its probability density function (pdf), p(x), assigns density to each possible value x.

Probabilities are areas under the density. Here a and b are the interval endpoints.

$$\Pr(a < X \leq b) = \int_a^b p(x)\,\mathrm{d}x$$



Red: density p(x).
Blue: cumulative probability P(x) = Pr(X ≤ x).
Green: interval width δx.

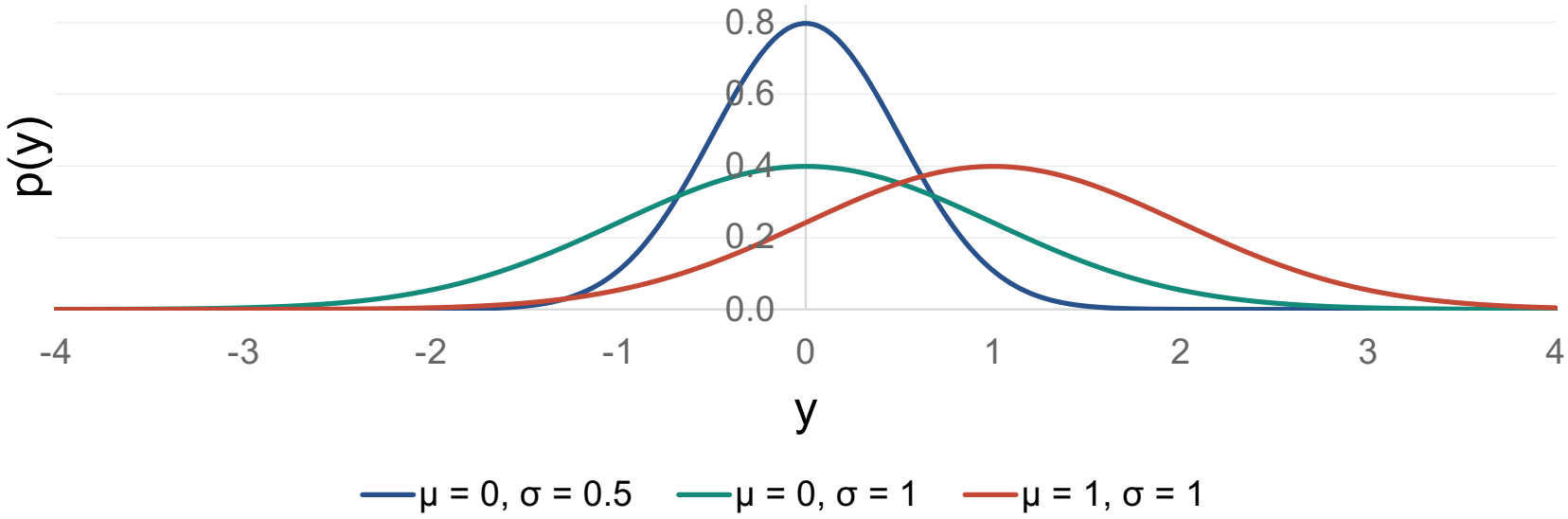
$$p(x) \geq 0, \qquad \int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = 1$$

Pr(X = x) = 0 at a single point. Density values can exceed 1.

The one-dimensional Gaussian distribution

$$Y \sim \mathcal{N}(\mu, \sigma^2), \quad p(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$$

∼: distributed as. N(mean, variance): Gaussian distribution.



μ: mean. σ: standard deviation. The variance is σ^2.

Likelihood and the fitting objective

N observations x_n, independently drawn from one distribution.
θ is the model's parameter vector. Hold the data fixed and vary θ.

Likelihood L

$$L(\boldsymbol{\theta}) = \prod_{n=1}^{N} p(x_n; \boldsymbol{\theta})$$

Log-likelihood: a product becomes a sum

$$\log L(\boldsymbol{\theta}) = \sum_{n=1}^{N} \log p(x_n; \boldsymbol{\theta})$$

Fitted parameters minimise negative log-likelihood C

$$C(\boldsymbol{\theta}) = -\log L(\boldsymbol{\theta}), \qquad \hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} C(\boldsymbol{\theta})$$

Gaussian maximum likelihood estimates

Fit a Gaussian to N scalar observations.

Negative log-likelihood

$$C(\mu, \sigma^2) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2$$

Set the mean derivative to zero

$$\frac{\partial C}{\partial \mu} = \frac{N\mu - \sum_{n=1}^{N} x_n}{\sigma^2} = 0$$

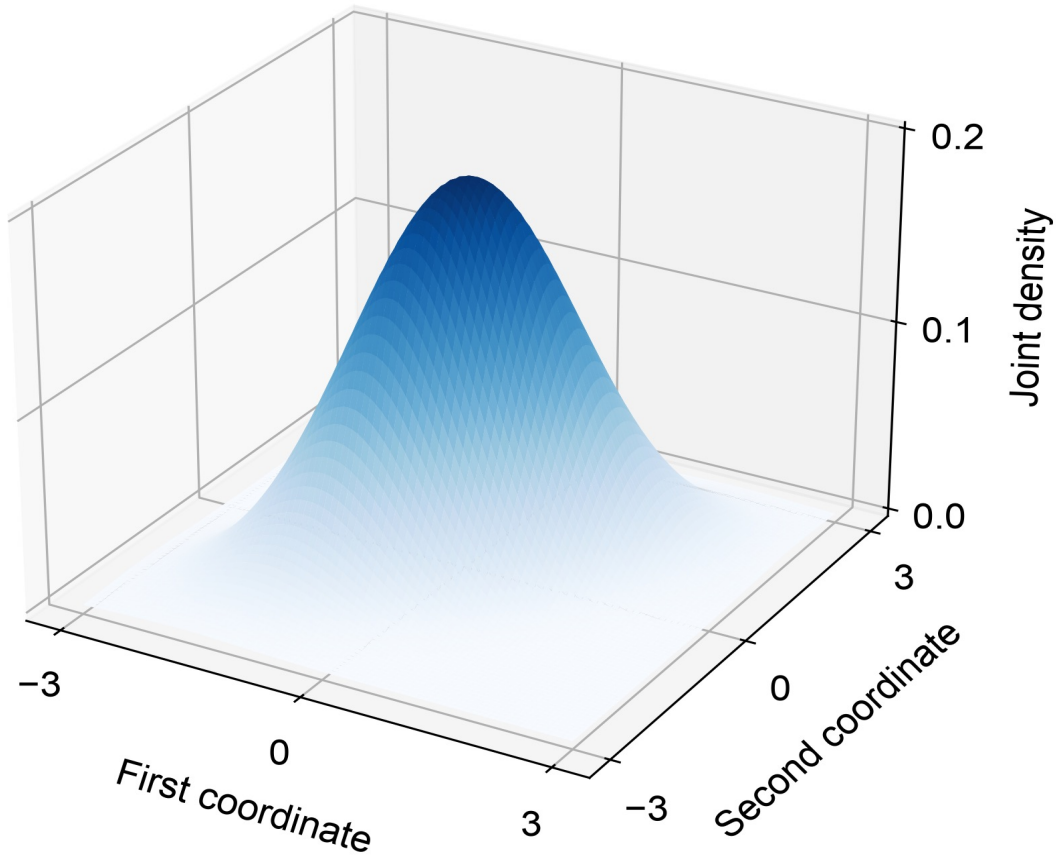
Sample mean

$$\hat{\mu} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

Sample variance (MLE uses N)

$$\hat{\sigma}^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \hat{\mu})^2$$

A Gaussian density in two dimensions



Two coordinates describe one outcome.

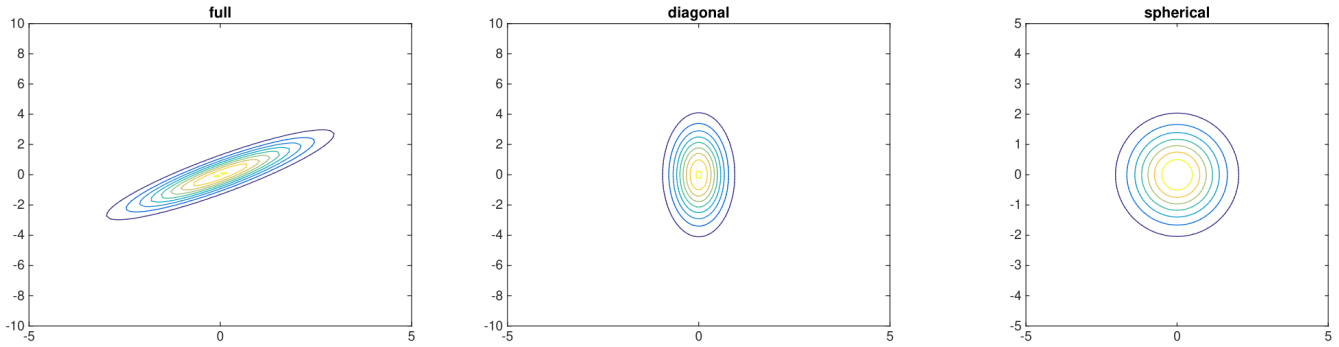
The mean sets the centre. Covariance controls spread and orientation.

Probability is volume under the surface over a region of the plane.

The multivariate Gaussian distribution

$$\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{Y})$$

Y has d coordinates. μ is the mean vector. Σ is the covariance matrix.



Diagonal entries: variances
Off-diagonal entries: how coordinates vary together

Diagonal covariance and sums of squares

$$\boldsymbol{\Sigma} = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_d^2)$$

Diagonal Σ: σ_j^2 is coordinate j's variance. Σ^{-1} is its inverse.

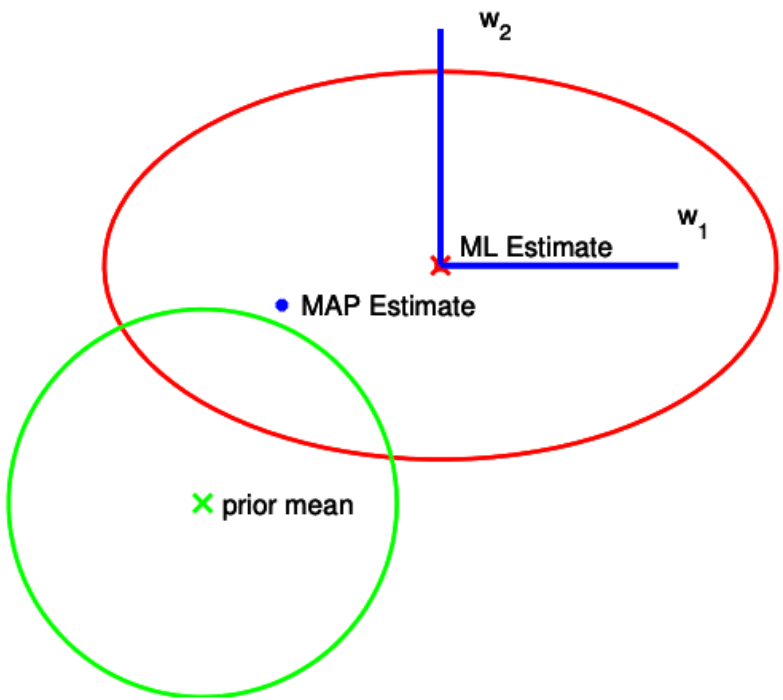
$$(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \sum_{j=1}^{d} \frac{(y_j - \mu_j)^2}{\sigma_j^2}$$

Fixed diagonal variances give a variance-weighted sum of squares.

$$\boldsymbol{\Sigma} = \sigma^2 \mathbf{I} \quad \Longrightarrow \quad \frac{1}{\sigma^2} \sum_{j=1}^{d} (y_j - \mu_j)^2 = \frac{1}{\sigma^2} \|\mathbf{y} - \boldsymbol{\mu}\|_2^2$$

Fixed equal variances give ordinary least squares, up to scale.

Ridge regression as MAP estimation



Likelihood

Scores how well parameters explain the observed data.

Prior

Describes parameter beliefs before these observations.

Posterior

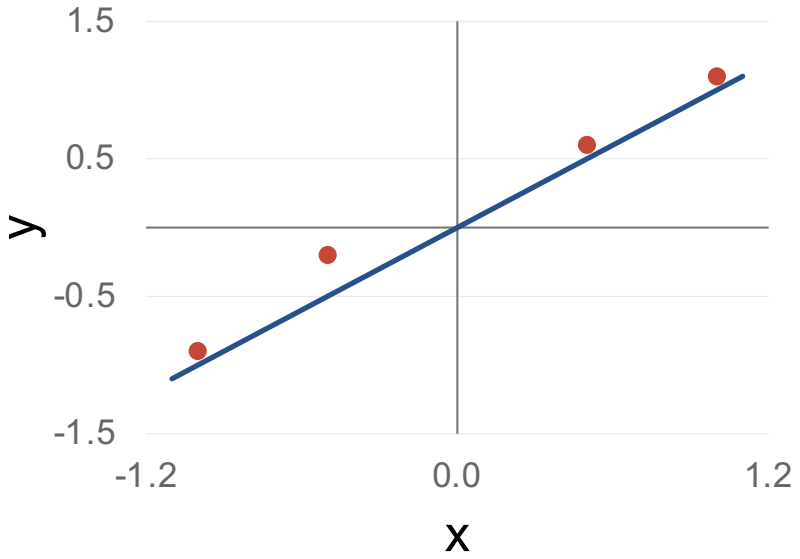
Combines likelihood and prior after observing the data.

w_1, w_2: two coefficients. ML: maximum likelihood.
MAP: maximum a posteriori. A prior gives a probabilistic basis for a penalty.

A probabilistic model for regression

$$y_i = \mathbf{x}_i^T \mathbf{w} + \epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$



ε_i: independent observation noise
σ^2: noise variance

$$p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \mathbf{x}_i^T \mathbf{w})^2}{2\sigma^2}\right]$$

Density at the observed target y_i.
Fix the data and vary w: a likelihood contribution.

Rasmussen & Williams (2006), §2.1.1

Gaussian likelihood gives least squares

$$-\log p(\mathbf{y} \mid X, \mathbf{w}, \sigma^2)$$

$$= \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|\mathbf{y} - X\mathbf{w}\|^2}{2\sigma^2}$$

For fixed σ², constants and positive scaling do not change the minimiser.

ML: maximum likelihood. Fit the coefficients by minimising squared error.

$$\hat{\mathbf{w}}_{\mathrm{ML}} = \arg\min_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|^2$$

A Gaussian prior will add the ridge penalty.

A Gaussian prior on the coefficients

Centred data. Fit the intercept separately without a penalty.

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 I)$$

τ^2: prior coefficient variance. I: identity matrix.

A smaller τ^2 puts more prior mass near zero.

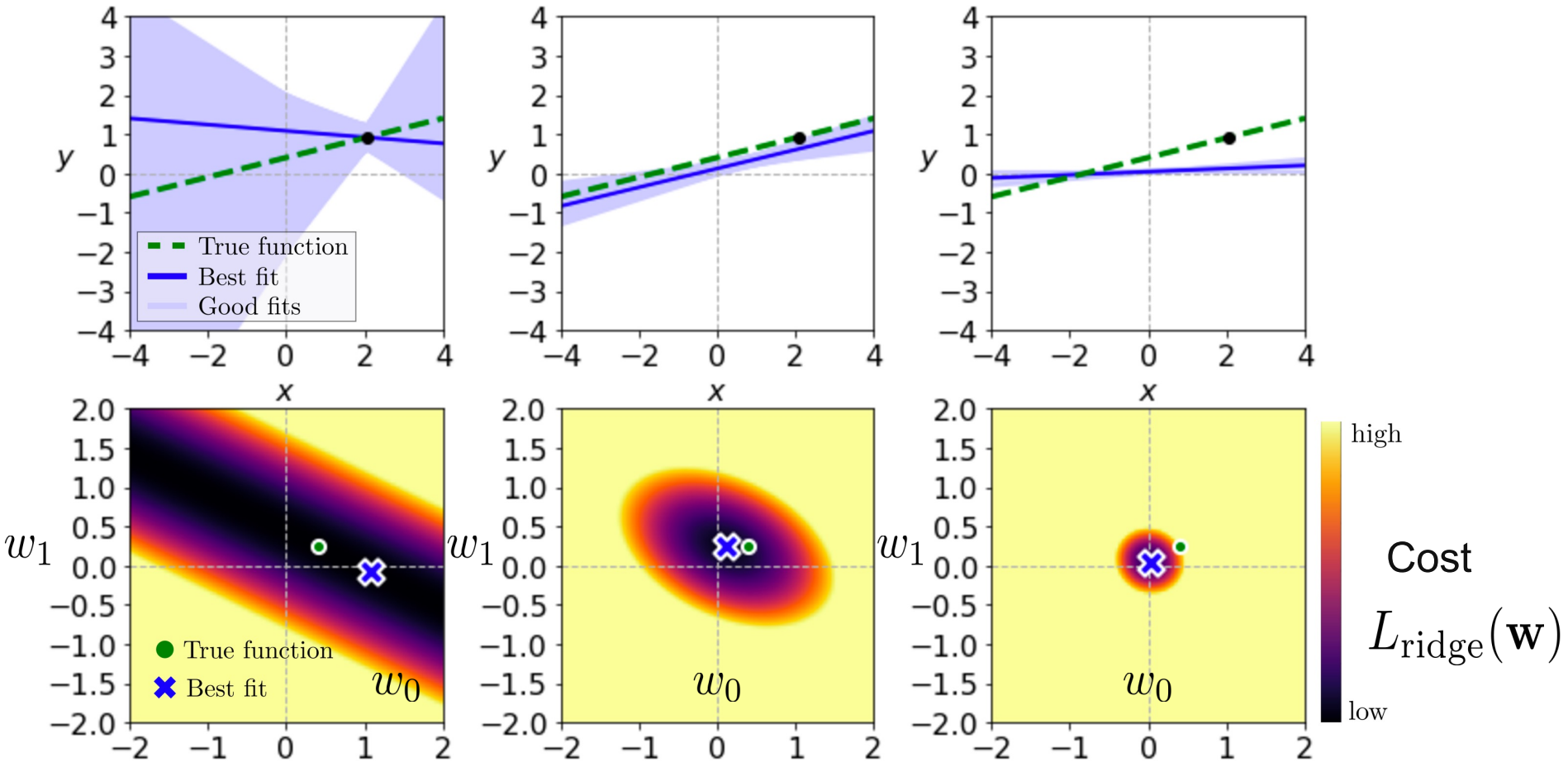
$$-\log p(\mathbf{w}) = \frac{\|\mathbf{w}\|^2}{2\tau^2} + \text{constant}$$

Rasmussen & Williams (2006), §2.1.1

Monday: regularisation restricts good fits

$$y = w_0 + w_1 x$$

Stronger penalty from left to right



This schematic penalises both coefficients. Our regression objective leaves the intercept unpenalised.

MAP estimation gives the ridge objective

MAP selects the coefficients with highest posterior density.

$$\hat{\mathbf{w}}_{\mathrm{MAP}} = \arg\max_{\mathbf{w}} p(\mathbf{w} \mid X, \mathbf{y})$$

Bayes' rule, then take negative logs

$$= \arg\min_{\mathbf{w}} \left\{ \frac{\|\mathbf{y} - X\mathbf{w}\|^2}{2\sigma^2} + \frac{\|\mathbf{w}\|^2}{2\tau^2} \right\}$$

Discard constants and multiply by $2\sigma^2$

$$= \arg\min_{\mathbf{w}} \left\{ \|\mathbf{y} - X\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2 \right\}, \quad \lambda = \frac{\sigma^2}{\tau^2}$$

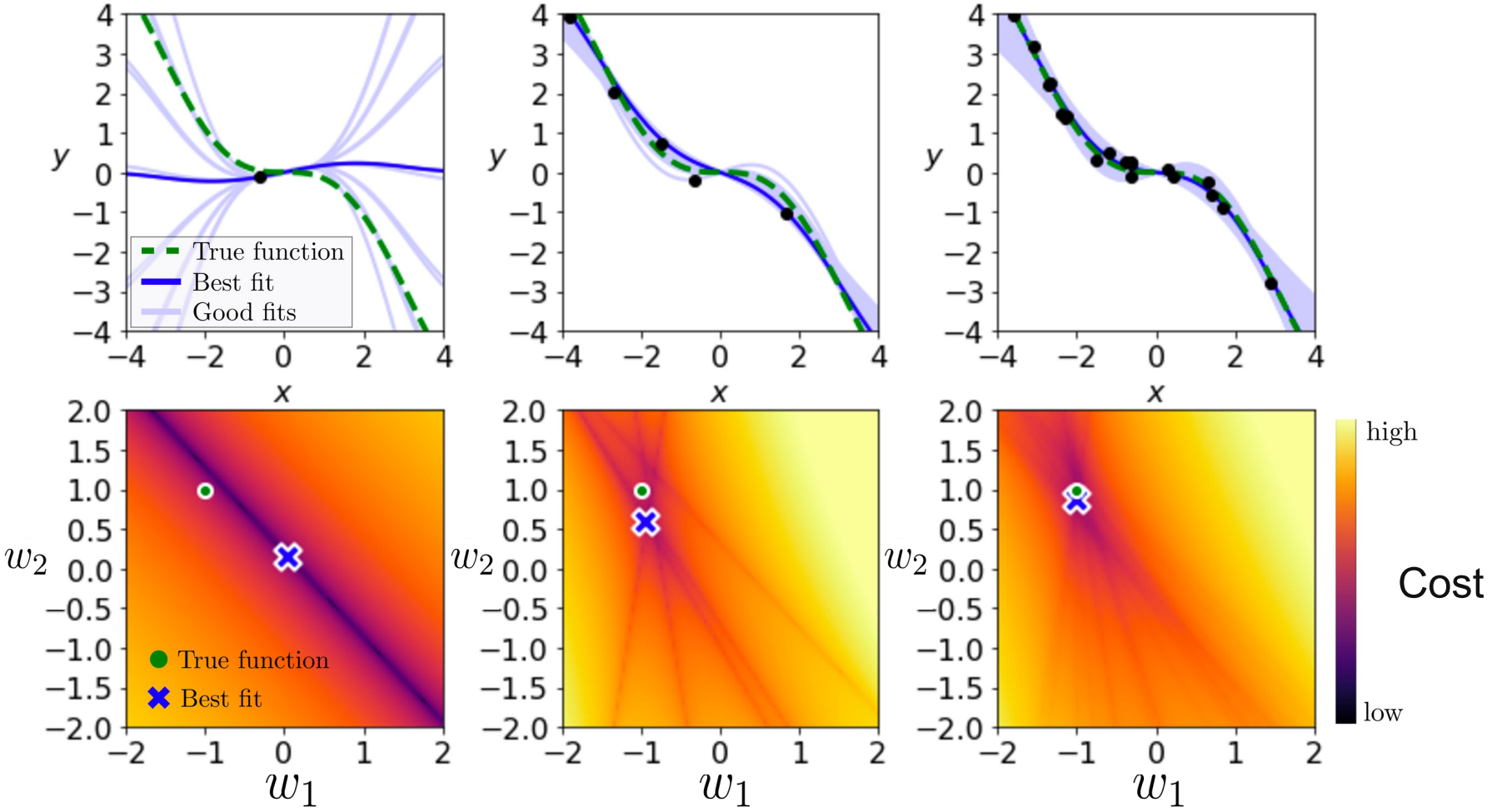
For mean squared error (RSS / N):

$$\lambda_{\mathrm{MSE}} = \frac{\sigma^2}{N\tau^2}$$

Monday: more observations constrain fitting

$$y = w_1 x + w_2 \sin(x)$$

More data from left to right



w_1, w_2: coefficients. The function class stays fixed.
Illustrated cost: residual magnitude to the power 0.25.

Torralba, Isola & Freeman (2024), Foundations of Computer Vision, Fig. 11.5

Posterior uncertainty about the parameters

$$p(\mathbf{w} \mid X, \mathbf{y}) = \mathcal{N}(\mathbf{m}, S)$$

m: posterior mean. S: posterior covariance.

$$\mathbf{m} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$$

$$S = \left(\frac{X^T X}{\sigma^2} + \frac{I}{\tau^2} \right)^{-1}$$

More informative measurements reduce uncertainty

$$X^T X = NqI, \quad q > 0$$

Example: N measurements
q > 0: equal feature energy in every direction

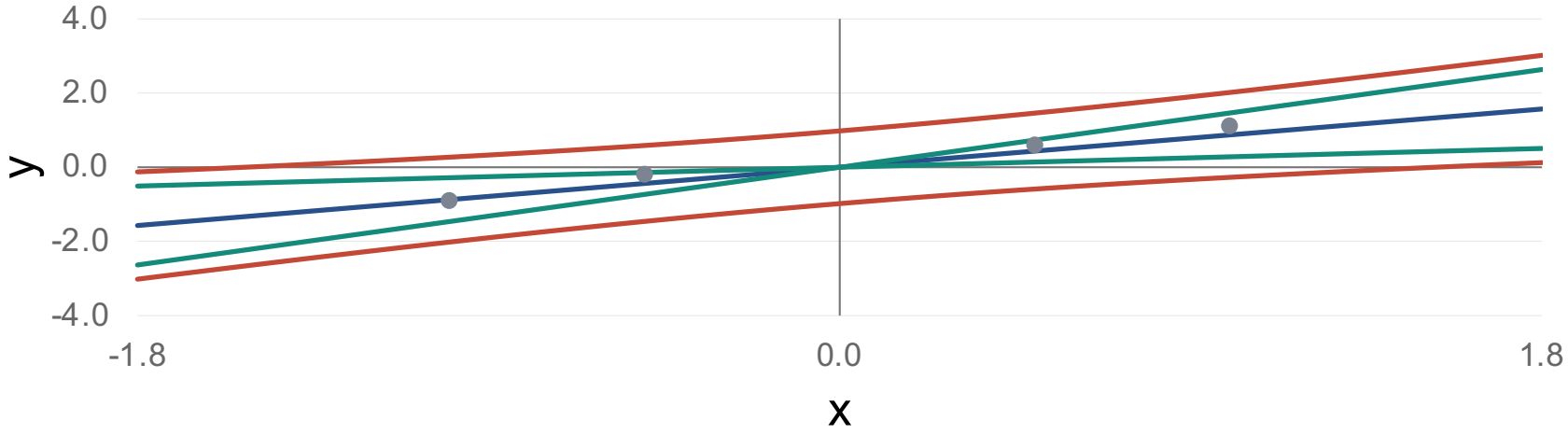
$$S = \left(\frac{Nq}{\sigma^2} + \frac{1}{\tau^2} \right)^{-1} I \longrightarrow 0 \quad (N \to \infty)$$

Mean uncertainty and observation noise

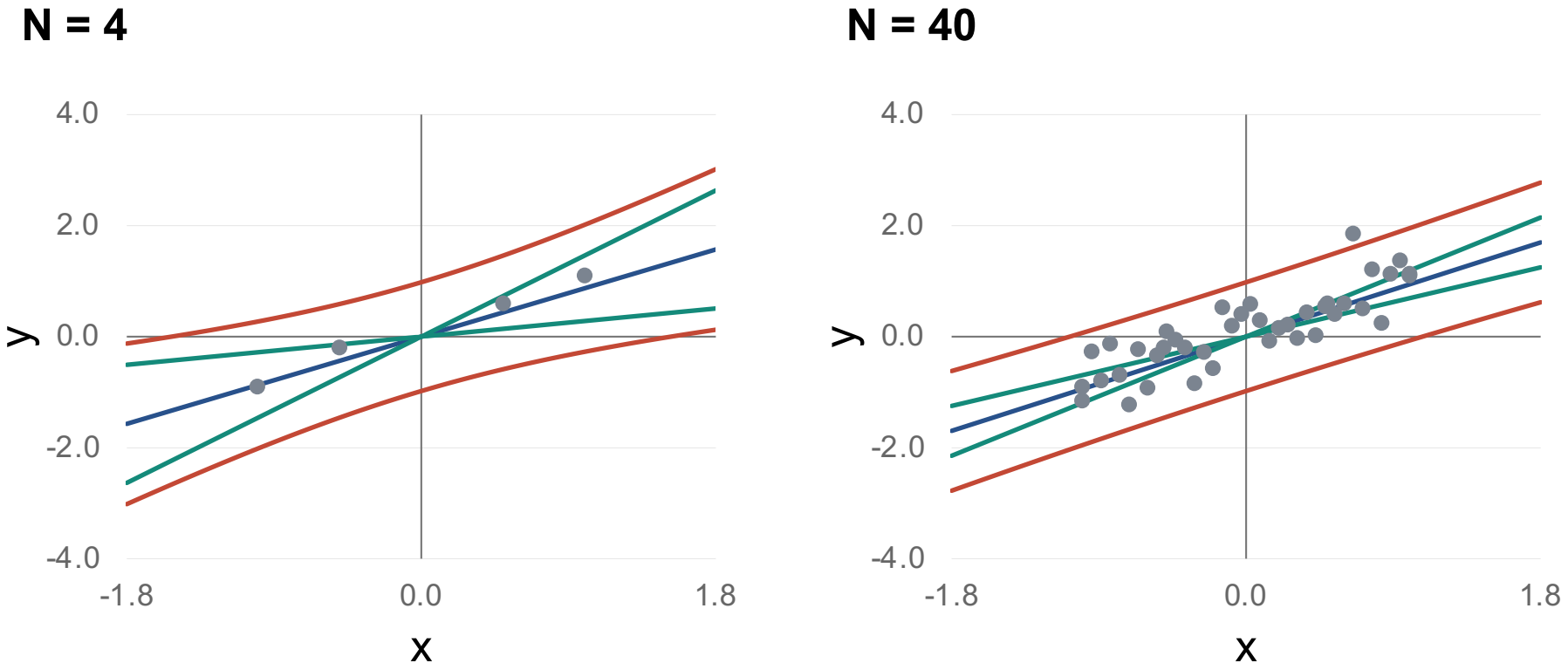
At new input x_*: f_* is the mean response; y_* is a noisy target.
D is the observed training set. Green: mean bounds. Red: new-observation bounds.

$$\mathrm{Var}(f_* \mid \mathcal{D}) = \mathbf{x}_*^T S \mathbf{x}_*$$

$$\mathrm{Var}(y_* \mid \mathcal{D}) = \mathbf{x}_*^T S \mathbf{x}_* + \sigma^2$$



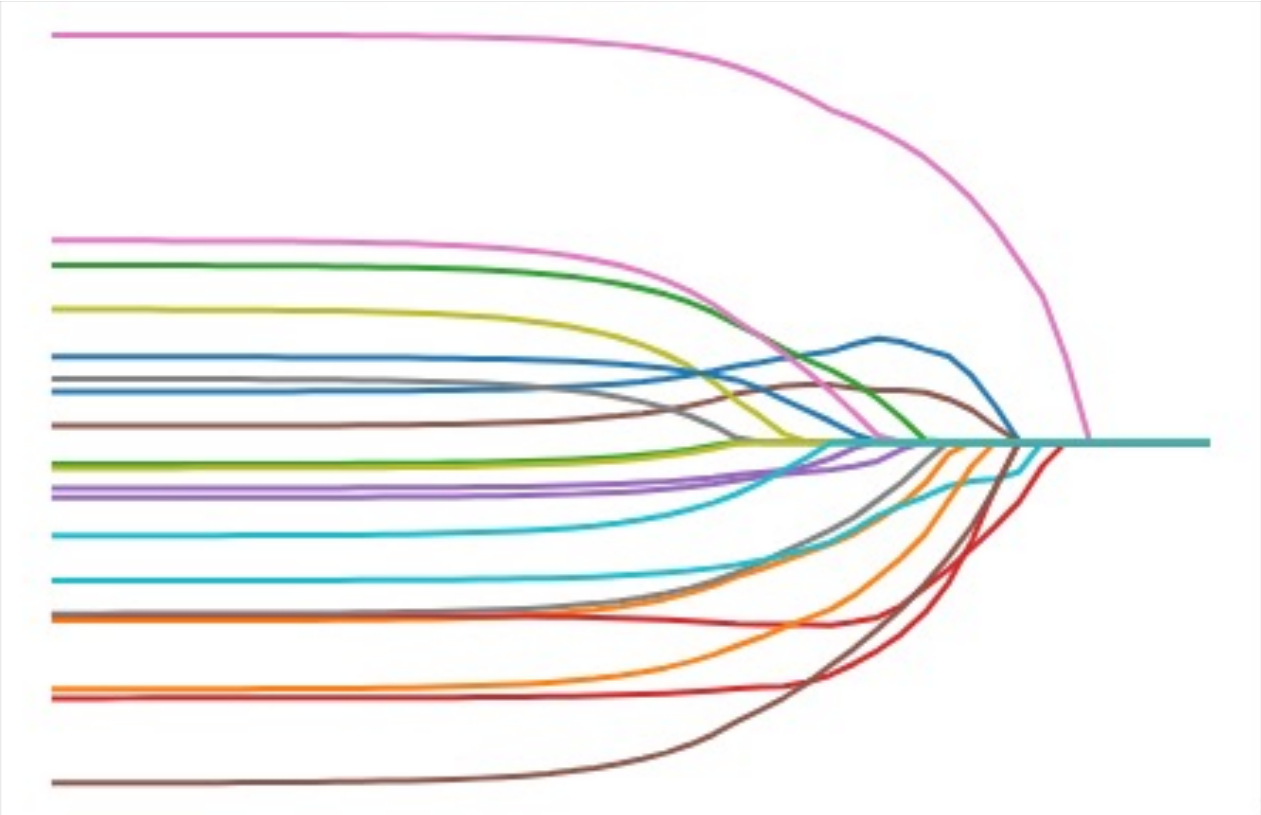
More data reduces parameter uncertainty



The green bounds narrow as the coefficient becomes better known.

The red bounds retain the contribution of observation noise.

L1 regularisation and LASSO

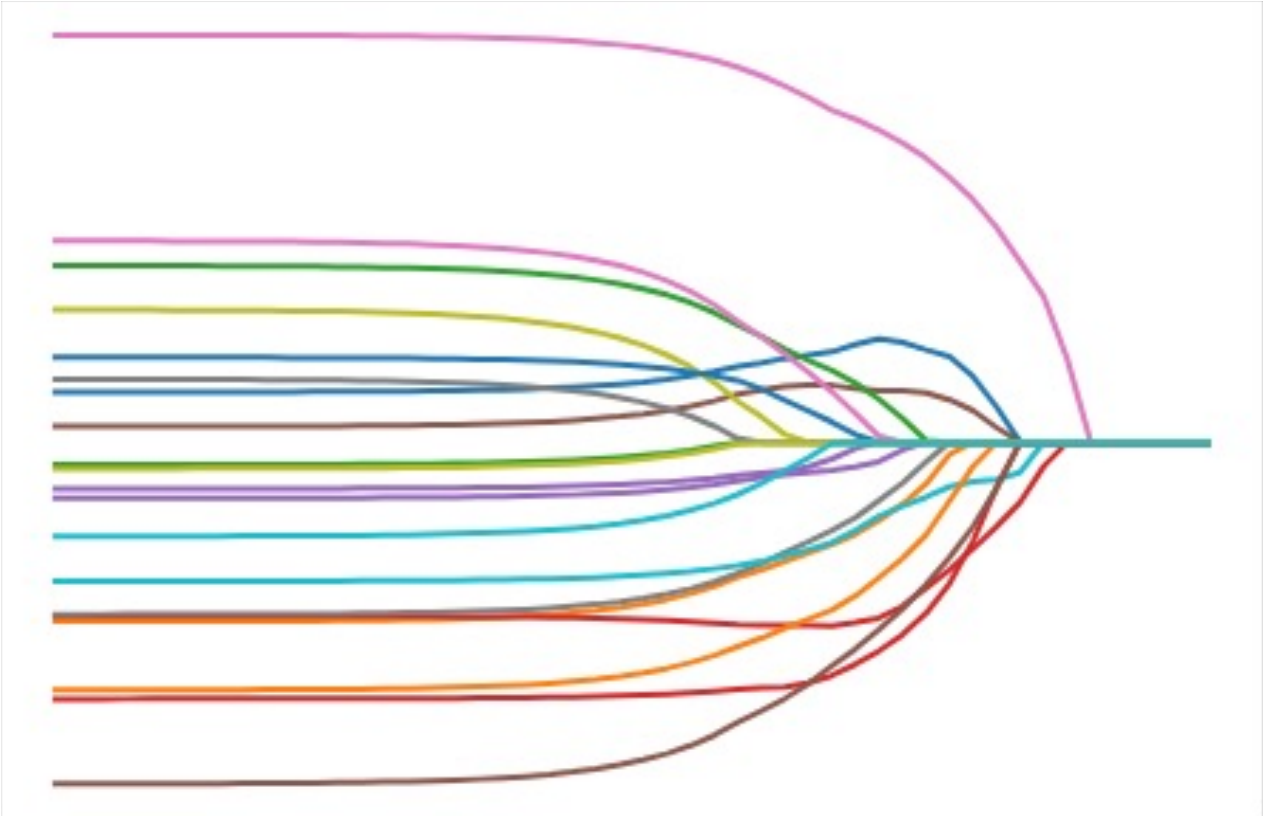


LASSO: least absolute shrinkage and selection operator

Tibshirani (1996), JRSS B

L1 regularisation and LASSO

The L1 penalty sums absolute coefficient values.



LASSO: least absolute shrinkage and selection operator

Tibshirani (1996), JRSS B

LASSO: an L1 coefficient penalty

Fit the data while penalising the sum of absolute coefficient values.

$$J_{\text{lasso}}(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|_1$$

J: penalised fitting cost. D: number of non-intercept coefficients.

$$\|\mathbf{w}\|_1 = \sum_{j=1}^{D} |w_j|$$

L1 norm

Some coefficients can become exactly zero.
LASSO selects a subset of the features.

Tibshirani (1996), JRSS B

Regularisation penalties and constraints

LASSO penalises L1. Ridge penalises L2 squared.

Penalty formulation

$$\min_{\mathbf{w}} \ \|\mathbf{y} - X\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_1$$

Constraint formulation

$$\min_{\mathbf{w}} \ \|\mathbf{y} - X\mathbf{w}\|_2^2 \quad \text{s.t.} \quad \|\mathbf{w}\|_1 \leq r$$

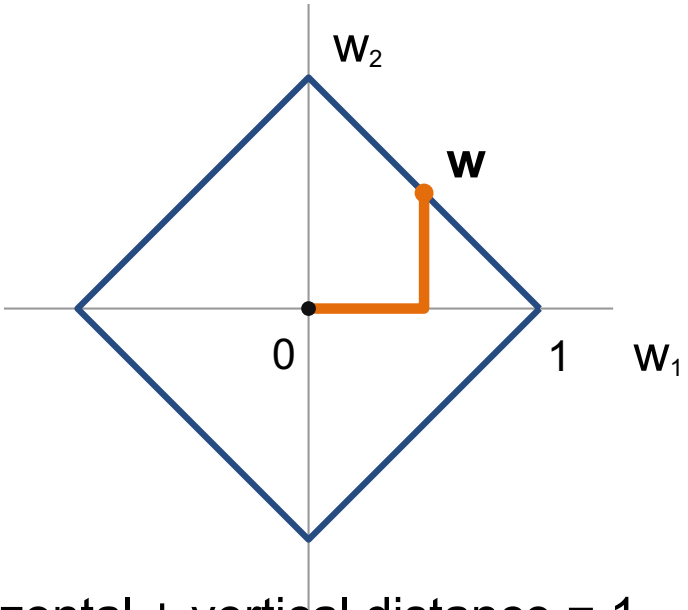
r: a matching radius. s.t.: subject to.
Next figures: α denotes w, and μ denotes the radius r.

Unit balls: L1 and L2 norms

w contains D coefficients. j indexes them.

L1: Manhattan distance

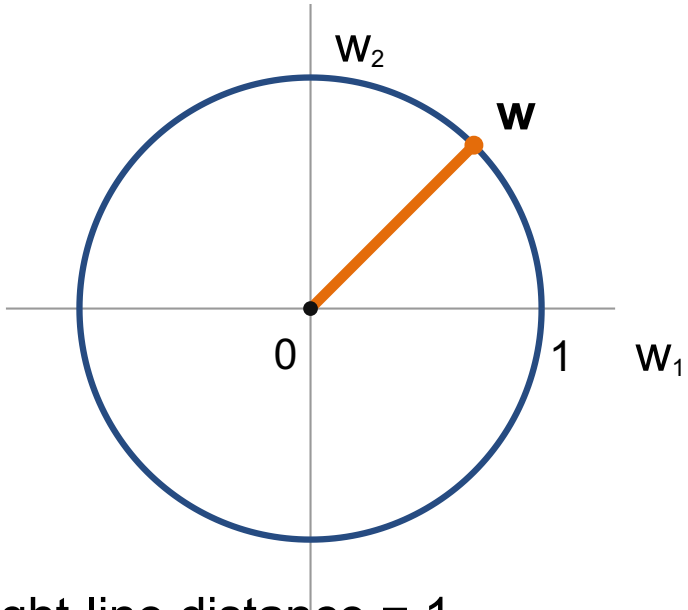
$$\|\mathbf{w}\|_1 = \sum_{j=1}^{D} |w_j|$$



Horizontal + vertical distance = 1

L2: Euclidean distance

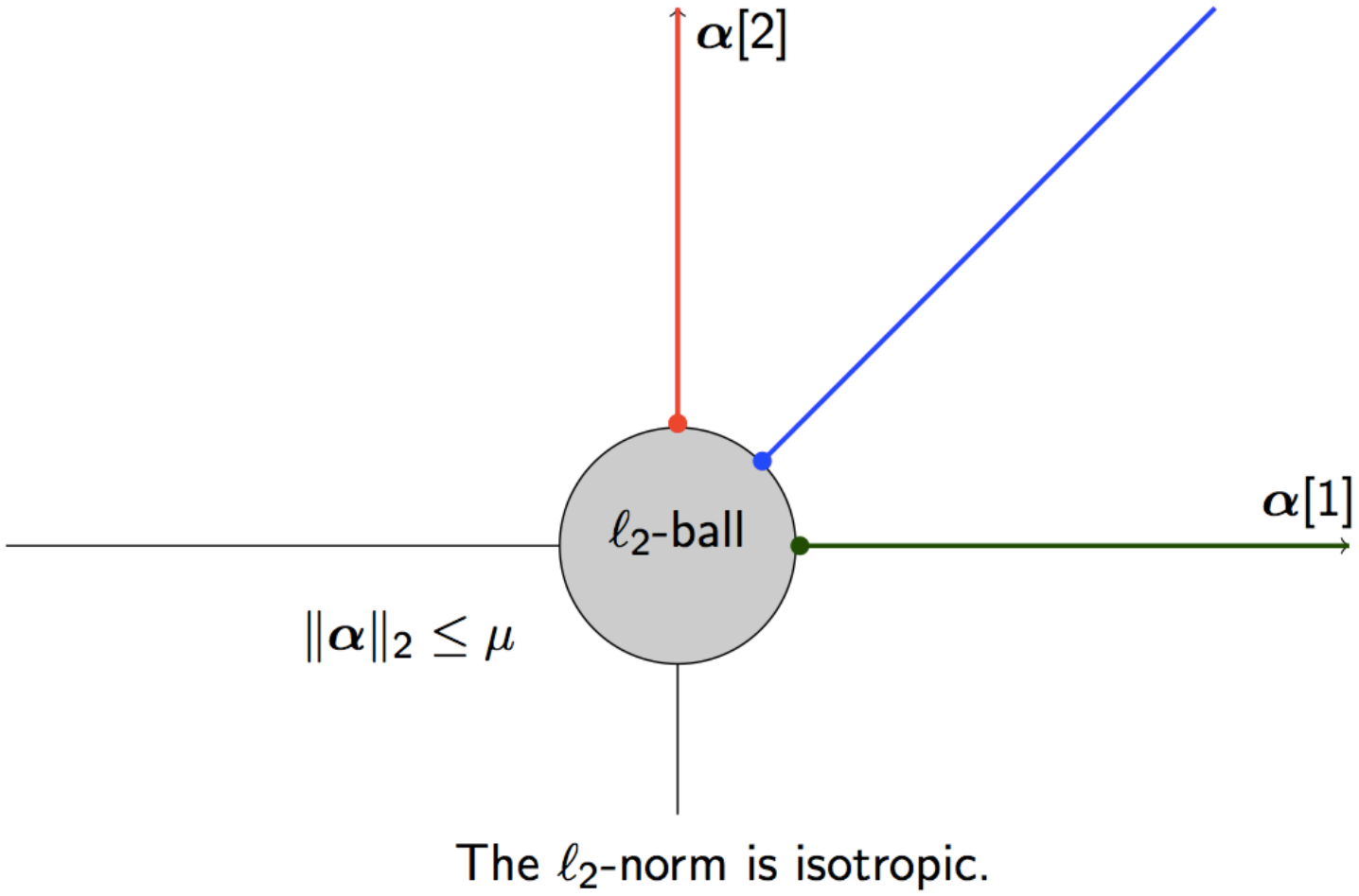
$$\|\mathbf{w}\|_2 = \left(\sum_{j=1}^{D} w_j^2 \right)^{1/2}$$



Straight-line distance = 1

Boundary: norm = 1. Unit ball: norm ≤ 1.

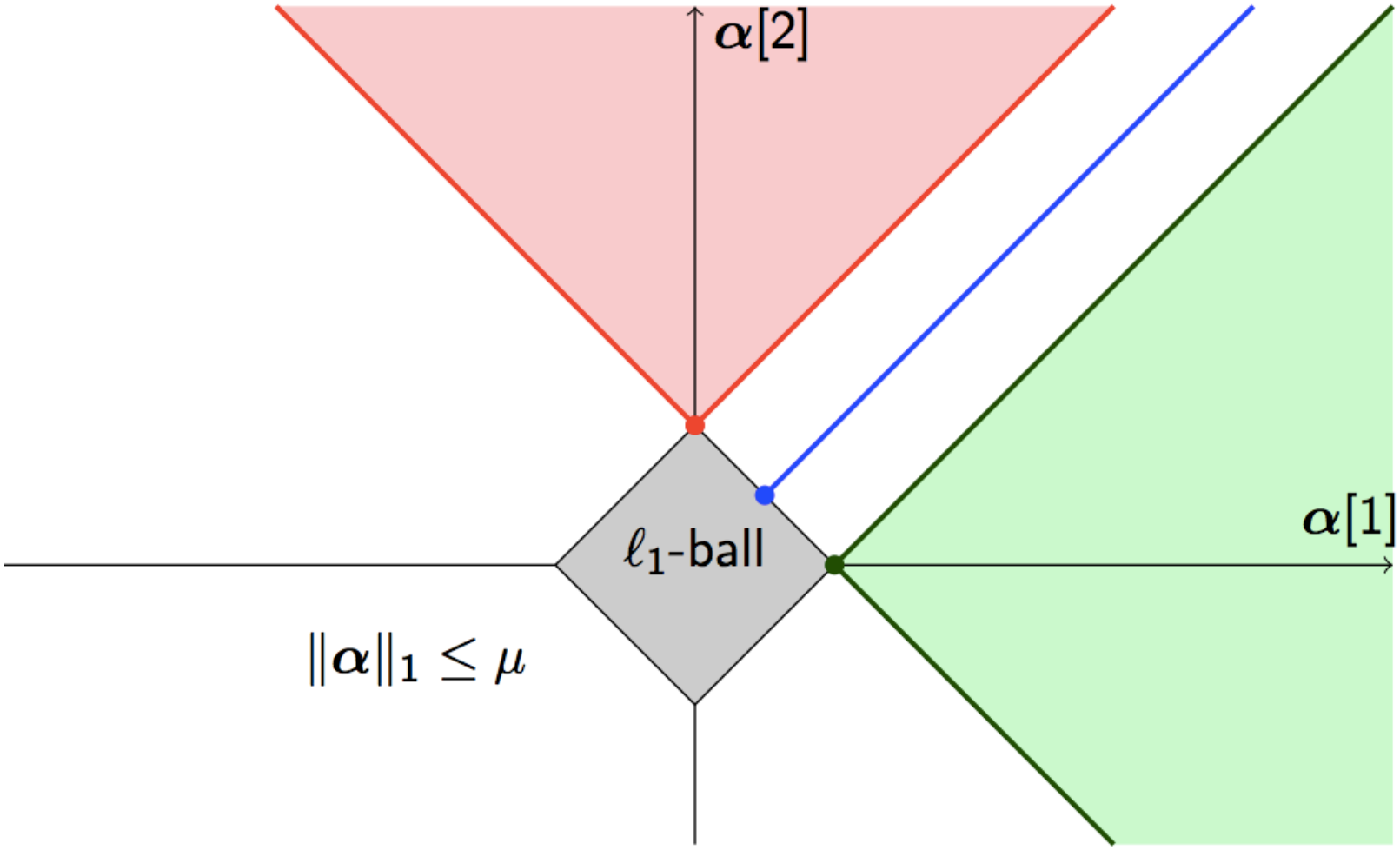
L2 geometry: projection onto a ball



The ℓ_2-norm is isotropic.

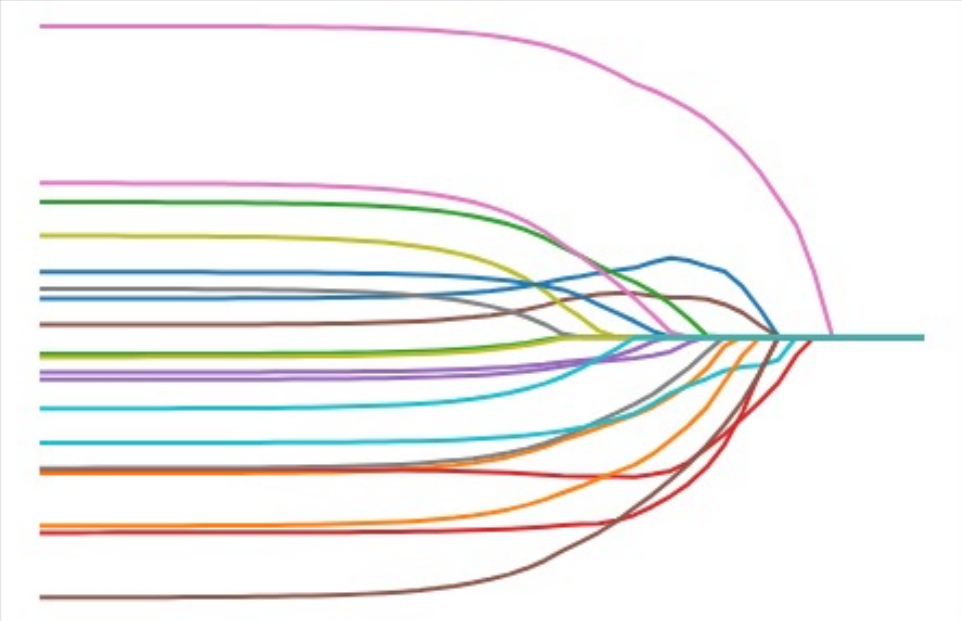
The nearest point usually has several nonzero coordinates.

L1 geometry: projection onto corners



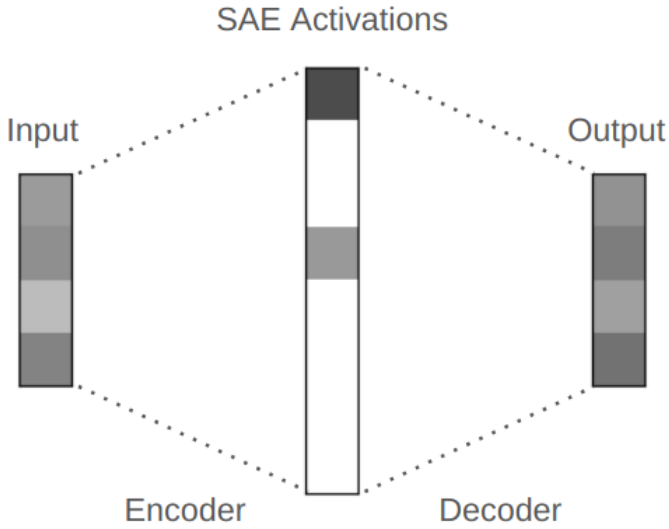
Many observations project onto a corner, where one coordinate is zero.

L1 norms: favors sparsity and interpretabiltiy



LASSO: least absolute shrinkage and selection operator

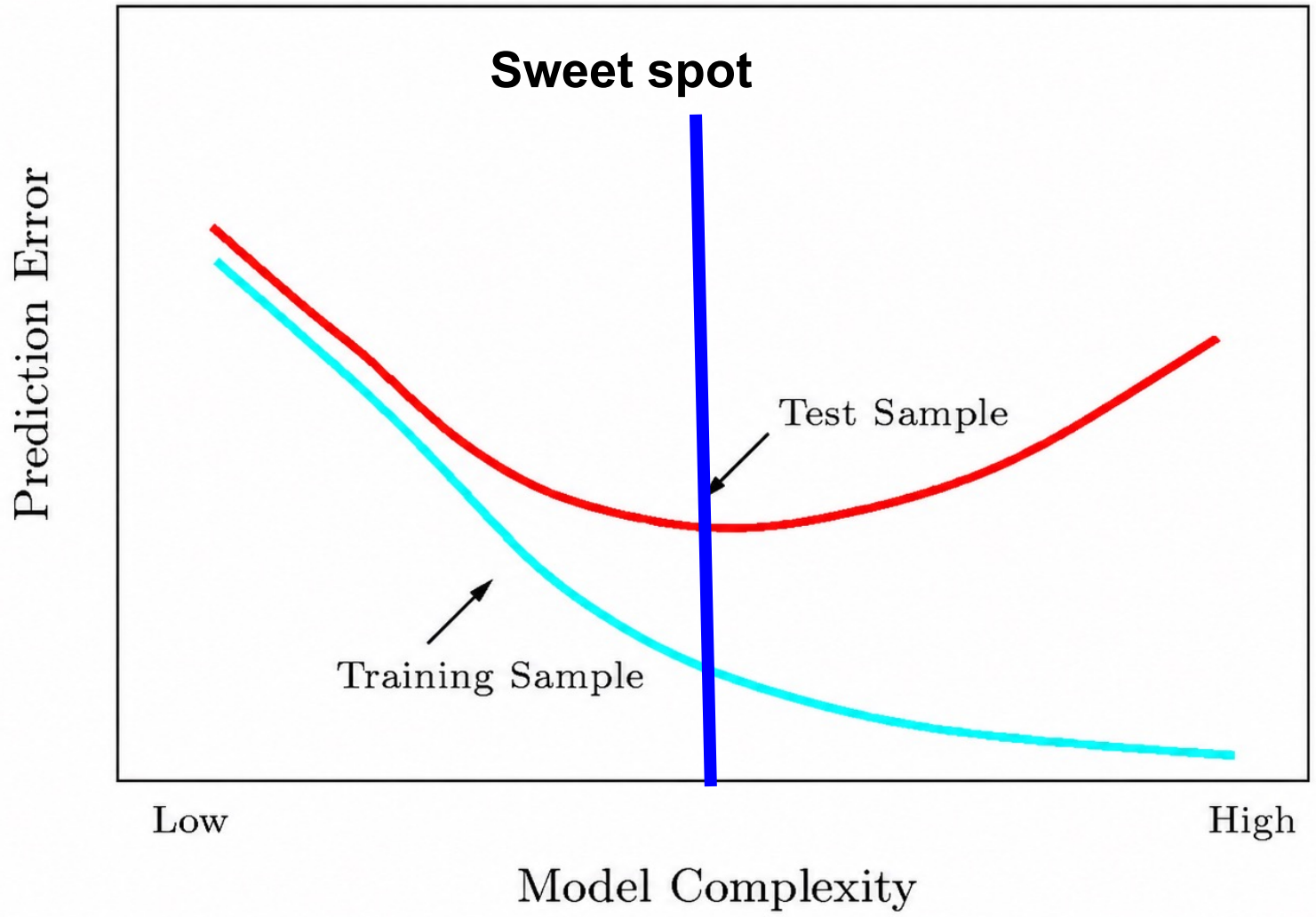
Tibshirani (1996), JRSS B



Sparse Autoencoders Find Highly Interpretable Features in Language M

https://arxiv.org/abs/2309.08600

Model complexity and held-out error



Larger ridge λ means lower complexity.

Expectation and a sample average

Z has density p(z). g(z) is a quantity computed from a value z.

Distribution: average over possible values

$$\mathbb{E}[g(Z)] = \int g(z)\, p(z)\, dz$$

Sample: estimate from N independent observations

$$\widehat{\mathbb{E}}[g(Z)] = \frac{1}{N} \sum_{i=1}^{N} g(z_i)$$

z_i: observations of Z. With g(z) = z, this is the sample mean.

Empirical risk minimisation

Training set D: N pairs with input x_i and target y_i, indexed by i.
f predicts a target; $\hat{y}$ is a prediction. ℓ is the squared loss.

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N} \qquad \ell(y, \hat{y}) = (y - \hat{y})^2$$

Training risk: mean squared error on the training set

$$\hat{R}_{\text{train}}(f) = \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, f(\mathbf{x}_i))$$

ERM: choose a function with minimum training risk

$$\hat{f}_{\mathcal{D}} \in \arg\min_{f \in \mathcal{F}} \hat{R}_{\text{train}}(f)$$

F: candidate function class. arg min selects a minimiser.
Subscript D: the training set used for this fit.

Population risk: the quantity we care about

P: real-world distribution. (X, Y): a fresh pair, independent of training.
Hold the fitted function fixed. R(f) is its population risk.

$$R(f) = \mathbb{E}_{(\mathbf{X},Y)\sim P}\big[\ell(Y, f(\mathbf{X}))\big]$$

$$= \int\!\!\int \ell(y, f(\mathbf{x}))\, p(\mathbf{x}, y)\, \mathrm{d}y\, \mathrm{d}\mathbf{x}$$

p(x, y): joint density for continuous inputs and targets.
The integral averages loss over real-world cases.

The population distribution is unknown.
Validation data will estimate unseen performance.

Validation estimates unseen performance

Reserve fresh validation pairs from the same population.
V: validation sample. N_{val}: its size. Superscript val marks its values.

$$\mathcal{V} = \{(\mathbf{x}_i^{\text{val}}, y_i^{\text{val}})\}_{i=1}^{N_{\text{val}}}$$

Evaluate the trained predictor without refitting it

$$\hat{R}_{\text{val}}(\hat{f}_{\mathcal{D}}) = \frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \ell(y_i^{\text{val}}, \hat{f}_{\mathcal{D}}(\mathbf{x}_i^{\text{val}}))$$

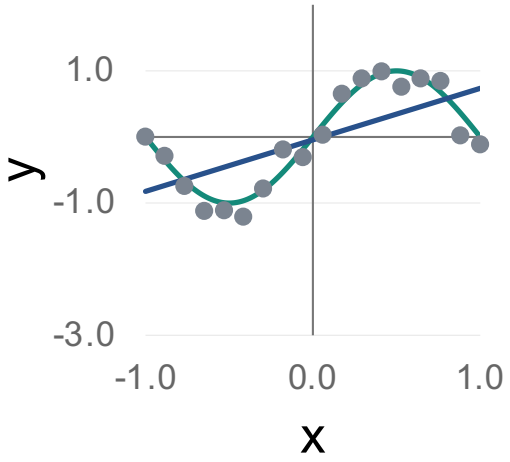
Hold training set D and the fitted model fixed:

$$\mathbb{E}_{\mathcal{V}}[\hat{R}_{\text{val}}(\hat{f}_{\mathcal{D}}) \mid \mathcal{D}] = R(\hat{f}_{\mathcal{D}})$$

E with subscript V averages validation samples, before model selection.
Use validation to choose complexity. Keep test data for final assessment.

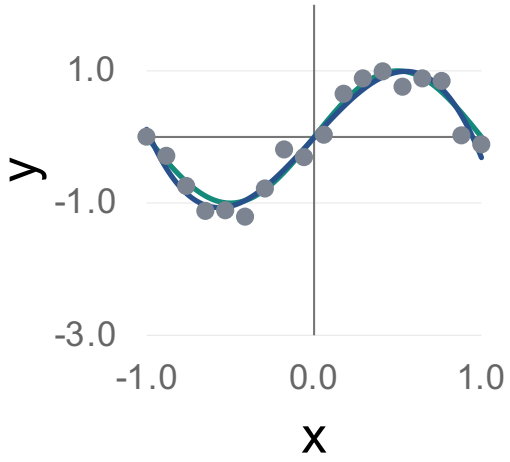
Training fit and unseen predictions

Degree 1



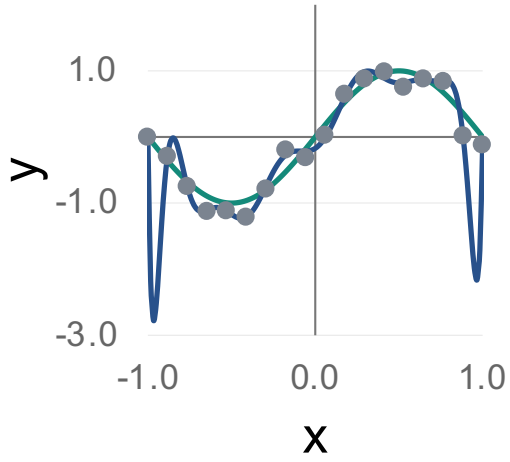
Train MSE: 0.305
Validation MSE: 0.245

Degree 3



Train MSE: 0.029
Validation MSE: 0.057

Degree 14



Train MSE: 0.002
Validation MSE: 0.485

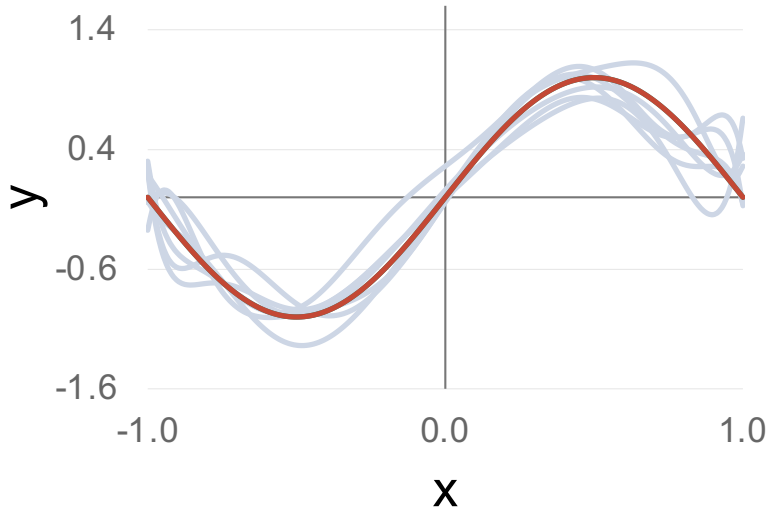
Same training set and 200 independent validation pairs for every model.
Grey points: training data. Blue: fitted predictor. Green: true mean.

The most flexible model wins on training error, but loses on validation.

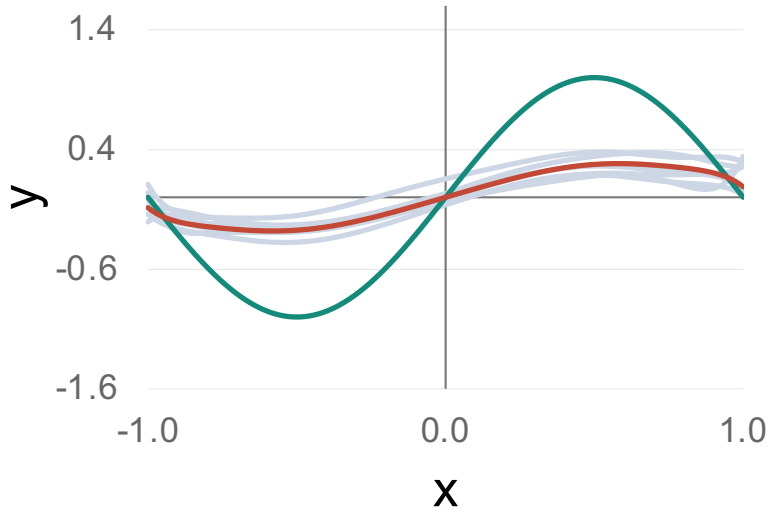
What does “averaging over datasets” mean?

Repeat collecting N examples from the same source and refitting.
Keep the function class and ridge strength λ fixed within each plot.

Weak ridge: λ = 0.0001



Strong ridge: λ = 10



At scalar input x: grey fits, red mean fit m(x), green true mean f*(x).

$$m(x) = \mathbb{E}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)], \qquad f^*(x) = \mathbb{E}[Y \mid X = x]$$

E with subscript D averages training sets.
Here: fixed inputs, fresh noisy targets. With random inputs, resample those too.

Bias–variance decomposition at one input

Fix x. Fresh target noise ε is independent of the training data.
σ²(x) is the noise variance at x. Keep the training method fixed.

$$Y = f^*(x) + \epsilon, \quad \mathbb{E}[\epsilon \mid x] = 0, \quad \mathbb{E}[\epsilon^2 \mid x] = \sigma^2(x)$$

Add and subtract the true mean and the mean fit

$$Y - \hat{f}_{\mathcal{D}}(x) = \underbrace{f^*(x) - m(x)}_{\text{mean error}} + \underbrace{m(x) - \hat{f}_{\mathcal{D}}(x)}_{\text{fit fluctuation}} + \epsilon$$

Square and average. Cross-terms vanish: the fluctuations have mean zero, and fresh noise is independent of the fitted predictor.

$$\mathbb{E}_{\mathcal{D}}[m(x) - \hat{f}_{\mathcal{D}}(x)] = 0, \qquad \mathbb{E}[\epsilon \mid x] = 0$$

$$\mathbb{E}_{\mathcal{D},Y}[(Y - \hat{f}_{\mathcal{D}}(x))^2 \mid X = x]$$

$$= \underbrace{(m(x) - f^*(x))^2}_{\text{bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(x) - m(x))^2]}_{\text{variance}} + \underbrace{\sigma^2(x)}_{\text{noise}}$$

Bias–variance explains population risk

Population risk averages fresh cases for one trained predictor.
Now also average over possible training sets D.

$$\mathbb{E}_{\mathcal{D}}[R(\hat{f}_{\mathcal{D}})] = \mathbb{E}_{\mathbf{X}}[\text{bias}^2(\mathbf{X}) + \text{variance}(\mathbf{X}) + \text{noise}(\mathbf{X})]$$

Average these terms over real-world inputs X.
This connects the fixed-input decomposition to population risk.

Validation estimates the risk of a particular fitted model

Bias–variance describes average risk as the training sample changes.
Validation estimates risk for the fitted model obtained from one sample.

A controlled ridge experiment

$$y_i = \sin(\pi x_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

18 fixed inputs across [−1, 1]. True mean: sin(πx).
Redraw independent Gaussian noise ε_i and refit on each new dataset.
σ is the noise standard deviation. Compare σ = 0.25 with σ = 1.

Degree-9 polynomial model, same fitting objective

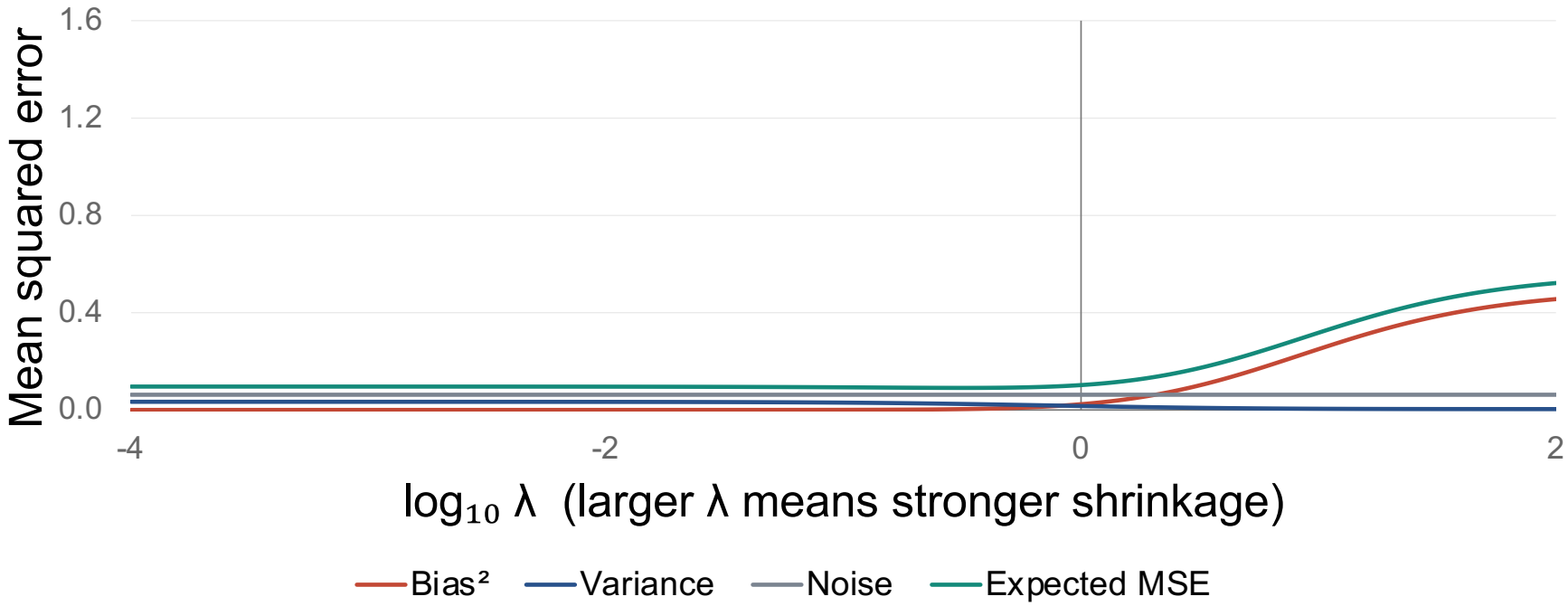
$$J(\mathbf{w}) = \|\mathbf{y} - \mathbf{\Phi w}\|_2^2 + \lambda \sum_{j=1}^{9} w_j^2$$

J: fitting cost. Φ: polynomial design matrix. y: target vector.
w: coefficients. λ: shrinkage strength. Intercept w_0 is unpenalised.

Curves average repeated datasets and fresh noise at 121 fixed inputs.

Ridge strength and expected error

Noise standard deviation σ = 0.25. All other conditions stay fixed.

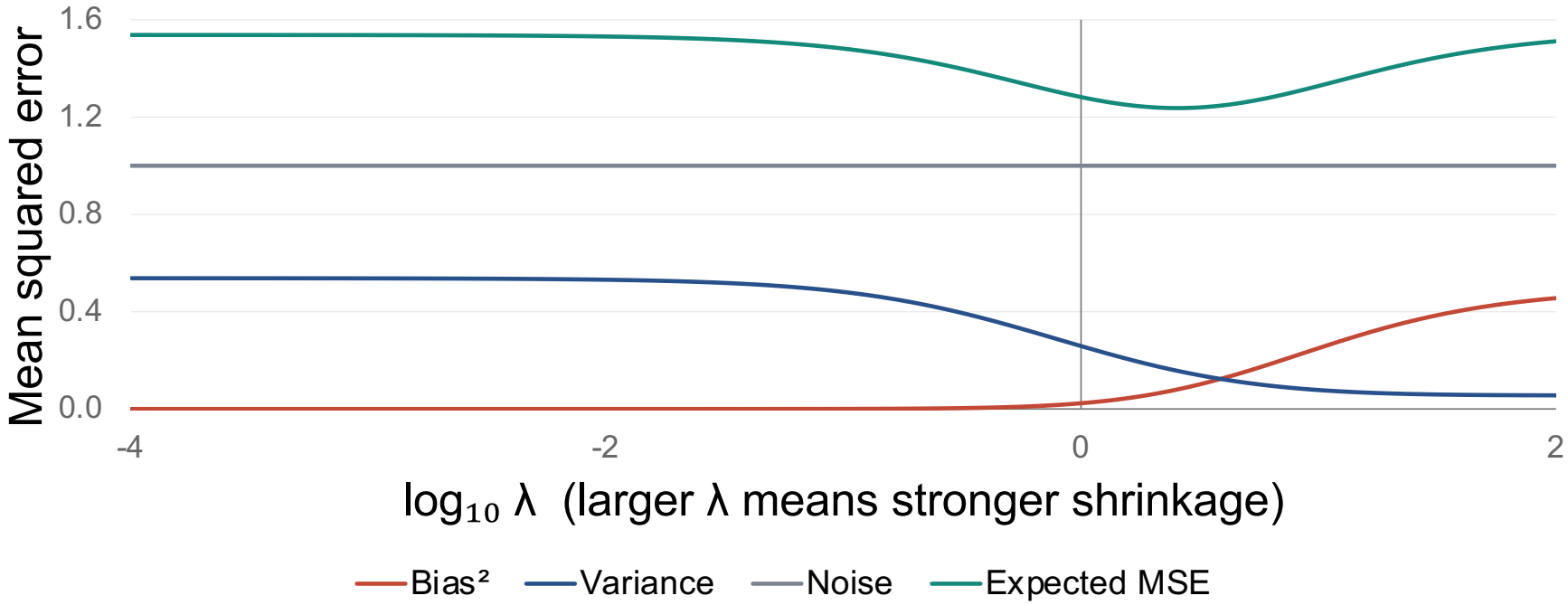


Minimum on this grid: λ ≈ 0.32, expected MSE ≈ 0.091.

Stronger shrinkage reduces variance but eventually adds too much bias.
An optimum need not lie in the interior for every problem.

More noise changes the ridge trade-off

Noise standard deviation σ = 1.00. All other conditions stay fixed.



Minimum on this grid: λ ≈ 2.51, expected MSE ≈ 1.237.

More noise raises the variance and noise floor. In practice, choose λ using validation. These curves use known simulated truth.

Symbol glossary: data and models

Data and dimensions

$\mathcal{D}$	Training dataset
N	Number of examples
i	Example index
j	Coordinate / feature index
D	Input dimension
M	Number of features
q	Polynomial degree
$\mathbf{x}_i$	Input vector for example i
y_i	Target for example i
$\mathbf{y}$	Vector of observed targets
$\mathbf{X}_i,\ Y_i$	Random input / target

Models and representations

$\mathbf{w}$	Parameter vector
w_0	Intercept
x_{i0}	Intercept coordinate (= 1)
$\hat{\mathbf{w}}$	Fitted parameter vector
$\boldsymbol{\phi}(\mathbf{x})$	Feature vector
$\phi_j(\mathbf{x})$	Feature j
$\mathbf{X}$	Input design matrix
$\boldsymbol{\Phi}$	Feature design matrix
$f_{\mathbf{w}}$	Prediction function
$\hat{y}$	Predicted target
ϵ_i	Residual / observation noise

Symbol glossary: fitting and probability

Fitting and generalisation

ℓ	Loss for one prediction
$L,\ J$	Fitting objective
λ	Penalty strength
R	Population risk
$\hat{R}_{\text{train}}$	Training risk
$\hat{R}_{\text{val}}$	Validation risk
K	Cross-validation fold count
f^*	True conditional mean
$m(x)$	Mean fitted prediction

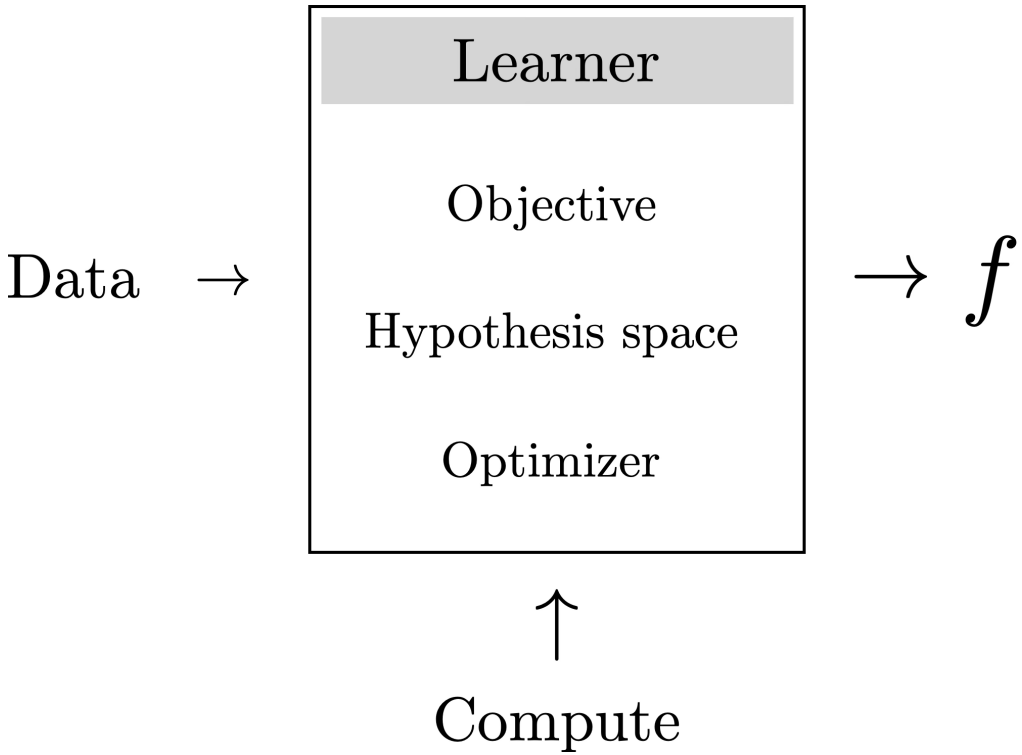
Local methods

k	Nearest-neighbour count
h	Kernel bandwidth
$\alpha_i(x)$	Weight of example i at x

Probability and uncertainty

P	Population distribution
p	Probability mass / density
$\mathbb{E}$	Expectation
$\mathrm{Var},\ \mathrm{Cov}$	Variance / covariance
$\mu,\ \boldsymbol{\mu}$	Scalar / vector mean
Σ	Gaussian covariance matrix
σ^2	Observation-noise variance
τ^2	Prior coefficient variance
$\mathbf{m}$	Posterior parameter mean
S	Posterior parameter covariance

Choosing a function by minimising cost



Objective

A cost scores task performance on the training set.

Function class

Candidates for prediction. f denotes one function. w denotes its parameter vector.

Optimisation

Choose parameters that minimise the training cost.

A reward measures desirability and is maximised.